Math 403 Homework #1, Spring 2025

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(1 point for each of up to 3 collaborators who also list you)

Due: 11:59pm Saturday 25 January 2025

Reading assignments (item numbers are lecture numbers as in PDF lecture notes) 52/57

1. for Thu. 9 January
   - [Hefferon, p.153–154] field axioms
   - [Cornell, "Fields" (about 1/3 of the way through `4330-week1.pdf`)]: more on fields
   - [Climenhaga, §7.2, §9.1] isomorphism, rank-nullity
2. for Tue. 14 January
   - [Lax, Chapter 1] quotients
   - [Climenhaga, §5.1] quotients
   - [Cornell, "Quotient Spaces" (`4330-week5.pdf`)] universal properties
   - [Cornell, "Exact Sequences" (about halfway through `4330-week5.pdf`)]
3. for Thu. 16 January
   - [Lax, Chapter 2] duality
   - [Climenhaga, §5.1, §8.2] duality, transpose
   - [Treil, §5.1–5.5] Hermitian inner products, ajoints; §5.2–5.4 should be mostly review
4. for Tue. 21 January
   - [Lax, Appendix 15] Jordan canonical form (very short proof)
   - [Hefferon, §5.IV.1] characteristic polynomial and minimal polynomial
   - [Treil, §9.3 and §9.5] generalized eigenspaces and Jordan canonical form
5. for Thu. 23 January
   - [Lax, Chapter 14, p.214–221] norms, equivalence, continuity, local compactness
   - [Stewart–Sun, §II.1] norms, equivalence, Hahn–Banach theorem
6. for Tue. 28 January
   - [Lax, Chap.12, p.187–190] Convex sets
7. for Thu. 30 January
   - [Serge Lang, *Linear Algebra*, Chapter XII, §1–§2] separating hyperplanes
   - [Serge Lang, *Linear Algebra*, Chapter XII, §3–§4] support hyperplane, extreme point

Exercises

1. (Freshman's Dream): The *characteristic* of a field F is the smallest positive integer p such that the sum $1 + \cdots + 1$ of p multiplicative identities is 0 in F. Prove that p is prime if it is finite. If F has characteristic p, show that $(a+b)^p = a^p + b^p$ for $a, b \in F$. 3/3

   Solution
   For the sake of notional ease take the symbol 2 to mean 1+1.

$$\begin{aligned}
&2 := 1 + 1 \\
&3 := 1 + 1 + 1 \\
&4 := 1 + 1 + 1 + 1 \\
&\dots \\
&p := 0 = 1 + 1 + 1 \dots
\end{aligned}$$

   From now on all the operations, identities, and all the inverses are from the field F.

   We first see that by using the axioms of a field, multiplication with this notation is like modulo multiplication.

   Armed with our notation, let's attack the question.

   : Everything up to and including this sentence is unnecessary.

   Assume F is finite and its characteristic is a composite then there are two integeres a, b such that $p = a * b$. Thne you can find two elements a', b' such they are 1 just added a and b times respectively.

   There's no reason to call a and b something other than what they are.

   But this also implies that 0 would have an inverse because $a' * b' * b'^{-1} = a' = 0 * b'^{-1} = 0$, so this implies $0 = b'^{-1}$. But this is obviously nonsense.

   So if F is finite p has to be prime.

   Going to the second question, we can use the binomial theorem to get that

$$(a' + b')^p = \sum_{k=0}^{p} \binom{p}{k} (a')^p (b')^{p-k} \tag{1}$$

   If we look at the terms with $k \neq 0$ and $k \neq p$, we get that $\binom{p}{k} = p * ((p-1)!/(p-k)!k!)$, since p=0, then all these terms evalute to 0 giving $(a' + b')^p = (a')^p + (b')^p$

2. Fix a vector space V and a subspace $W \subseteq V$ over a field F. Let $\pi : V \to V/W$ be the *projection homomorphism* given by $\pi(v) = v + W$. Write X for the set of all subspaces of V that contain W, and write Y for the set of all subspaces of V/W. Prove that π induces a bijection between these two sets, with 0/3

This is extremely hard to read.

$$X \to Y$$
$$L \mapsto \pi(L) = \{\pi(v) \mid v \in L\}$$

and

$$Y \to X$$
$$M \mapsto \pi^{-1}(M) = \{v \in V \mid \pi(v) \in M\}.$$

Solution

Let $L \in X$, then $W \subseteq L$.

Then $dim(L) \geq dim(W)$. So assume $dim(L) = x$ and $dim(W) = y$, $\Rightarrow x \geq y$. $\to$ $v_1, ..., v_x$ span L.

Change this vectors so that $v_1, ..., v_y$ span W and the $y - x$ other vectors span the complement of W in L. Now if you take the subspace spanned by $[v_{y+1}], [v_{y+2}], [v_{y+3}], ..., [v_x]$, they are a subspace of V/W, and this subspace and L are the equivalent(upto isomorphism), because the basis of this space is the basis for W plus the other $x - y$ vectors, which is the same as L.

This doesn't show surjectivity of π. All you've shown here is that the image of π is a vector space of the same dimension.

Take $M \in Y$, then it has a basis of say x vectors. If you take the union of the elements in the basis(one elment from the coset) and the basis for W, you get an element in Y. So π^{-1} is surjective.

This doesn't show surjectivity of π^{-1} and I'm not quite sure what you're saying. You have to start with something in X and construct something in Y mapping to it.

We have proved that π and π^{-1} are surjective, but we also need to prove that they are also injective.

For π, assume L_1 and L_2 both map to $M \in Y$. Now look at the basis for M, they are the basis for W plus v_1 up to v_y assuming $dim(M) = y$. Since $M = L_1$ and $M = L_2$, then $L_1 = L_2$ because they have the same set of basis, therefore are the same subspace.

More is needed for injectivity here.

For π^{-1}, assume M_1 and M_2, both map to $L \in Y$. Since $M_1 = L$ and $M_2 = L$, then $M_1 = M_2$, by the same reasoning as above. Henceforth, the function is bijective.

3. Show that giving an exact sequence $\cdots \to V_{i-1} \to V_i \to V_{i+1} \to \cdots$ is the same as giving a collection of short exact sequences $0 \to K_i \to V_i \to K_{i+1} \to 0$, one for each i. (The long exact sequence is said to be constructed by *splicing* the short exact sequences together.) 3/3

For the sake of simplicity, I will use the diagram below.

$$
\begin{tikzcd}
0 \arrow[dr] & & 0 & & 0 \arrow[dr] & & 0 & \\
& K_i \arrow[ur] \arrow[dr, hook] & & & & K_{i+2} \arrow[ur] \arrow[dr, hook] & & \\
\ldots \arrow[ur] & & V_i \arrow[dr, two heads] & & V_{i+1} \arrow[ur, two heads] & & V_{i+2} \arrow[dr, two heads] & \\
& & & K_{i+1} \arrow[ur, hook] \arrow[dr] & & & & \ldots \\
& & 0 \arrow[ur] & & 0 & & &
\end{tikzcd}
$$

I highlighted that the mapping between K_i and V_i is injective and that the mapping between V_i and K_{i+1} is surjective. Now consider what being an exact sequence entails, it just means that the you can make a mapping satisfying $img(\varphi_i) = ker(\varphi_{i+1})$. So if we can construct a mapping between the vector spaces that satisfies this property, then the aforementioned exact sequence does in fact exist.

First step let's make φ_i, such that its domain is V_i and its codomain is V_{i+1}. We can do this by first observing that there exists a surjective mapping between V_i and K_i. Next we can also observe that there is an injective mapping between K_i and V_{i+1}. If we compose these to mappings, we get a mapping whose domain is V_i and its codomain is V_{i+1}.

Next let's make φ_{i+1} whose domain is V_{i+1} and whose codomain is V_{i+2}, such that its kernel is the image of the previous mapping. We know that from the smaller chain consisting of V_{i+1} that there is a mapping from V_{i+1} to K_{i+2}, such that it sends all the elements that have a preimage in K_{i+1} to $\mathbf{0}$. We also now that there is an injective mapping between K_{i+2} and V_{i+2}. If we compose this two mappings, we get a mapping whose domain is V_{i+1} and whose codomain is V_{i+2}, such that its kernel is the image of K_{i+1} in V_{i+1}.

So now we have that the $img(\varphi_i) \subseteq ker(\varphi_{i+1})$, let's now prove that every element in the kernel of φ_{i+1} has a preimage in V_i, thereby completing the proof.

We know from the chain consisting of V_{i+1}, that the kernel of the mapping between V_{i+1} and K_{i+2}, is K_{i+1} by definition of exactness. We also now that each element in K_{i+1} has a preimage in V_i, so to conclude the kernel of the mapping between K_{i+2} and V_{i+2} is K_{i+1} and every element in K_{i+1} has a preimage in V_i. So $ker(\varphi_{i+1}) = img(\varphi_i)$. So given a collection of short sequences, you can make a longe chain. To do the reverse, take each the kernel of the homomorphism beteween V_i and V_{i+1} and define it as K_{i+1}. Now if you constrain homomorphism to this codomain, you get the short sequences again. Therefore a collection of short sequences is the same as one big long sequence.

4. Rank-nullity theorem for exact sequences: Given an exact sequence 3/3

$$0 \to V_0 \to V_1 \to \cdots \to V_r \to 0,$$

prove that $\sum_{i=0}^{r}(-1)^i \dim V_i = 0$.

Solution

For the sake of simplicity I will extend the exact sequence on the right by adding another **0**, and I will relabel the sequences as shown below.

$$0 \xrightarrow{\varphi_0} V_0 \xrightarrow{\varphi_1} V_1 \xrightarrow{\varphi_2} ... \xrightarrow{\varphi_r} V_r \xrightarrow{\varphi_{r+1}} 0 \xrightarrow{\varphi_{r+2}} 0$$

$$\begin{aligned} dim(V_i) &= img(\varphi_{i+1}) + ker(\varphi_{i+1}) \\ dim(V_i) &= ker(\varphi_{i+2}) + ker(\varphi_{i+1}) \end{aligned}$$

$$\begin{aligned} dim(V_0) &= ker(\varphi_2) + ker(\varphi_1) \\ (-1)dim(V_1) &= (-1)(ker(\varphi_3) + ker(\varphi_2)) \\ &... \\ (-1)^r dim(V_r) &= (-1)^r(ker(\varphi_{r+2}) + ker(\varphi_{r+1})) \\ \sum_i^r dim(V_i) &= ker(\varphi_1) + (-1)^r ker(\varphi_{r+2}) \end{aligned}$$

But $ker(\varphi_1) = 0$ and $ker(\varphi_{r+2}) = 0$, so their sum is 0, thereby completing the proof.

5. Rank-nullity for arbitrary complexes: Given a complex $0 \to V_0 \to V_1 \to \cdots \to V_r \to 0$, 3/3
write $B_i = \mathrm{im}(V_{i-1} \to V_i)$ and $Z_i = \ker(V_i \to V_{i+1})$. Prove that

$$\sum_{i=0}^{r}(-1)^i \dim V_i = \sum_i (-1)^i \dim H_i,$$

where $H_i = Z_i/B_i$ is the i^{th} homology of the complex. [In Exercise 4, $B_i = Z_i = K_i$, so $H_i = Z_i/B_i = 0$ for all i.]

Solution

From corollary of lecutre 2, $dim(H_i) = dim(Z_i) - dim(B_i)$.

$$\begin{aligned}\sum_{i=0}^{r}(-1)^i V_i = dim(Z_0) + dim(B_1) \\ - dim(Z_1) - dim(B_2) \\ \cdots \\ + (-1)^r(dim(Z_r) + dim(B_{r+1}))\end{aligned}$$

$$\begin{aligned}\sum_{i=0}^{r}(-1)^i H_i = dim(Z_0) - dim(B_0) \\ - dim(Z_1) + dim(B_1) \\ \cdots \\ + (-1)^r(dim(Z_r) - dim(B_r))\end{aligned}$$

$$\begin{aligned}\sum_{i=0}^{r}(-1)^i V_i - \sum_{i=0}^{r}(-1)^i H_i &= (dim(Z_0) - dim(Z_0)) + (dim(B_1) - dim(B_1)) + dim(B_0) \\ &+ (-dim(Z_1) + dim(Z_1)) + \cdots + dim(B_{r+1}) \\ &= dim(B_0) + dim(B_{r+1}) = 0 + 0 = 0\end{aligned}$$

So the sums are equal, because $dim(B_0)$ is 0 because 0 can only be mapped to 0 in V_1 and $dim(B_r)$=0 because V_r is mapped to 0, so their sum is 0.

6. Two elements u and v in a vector space V are *congruent modulo* a subspace $W \subseteq V$, 3/3 written $u \equiv v \pmod{W}$, if $u + W = v + W$. Show that congruence modulo W is an *equivalence relation* on V, meaning that it is

   - reflexive: $v \equiv v \pmod{W}$ for all $v \in V$;
   - symmetric: if $u \equiv v \pmod{W}$, then $v \equiv u \pmod{W}$; and
   - transitive: if $u \equiv v \pmod{W}$ and $v \equiv x \pmod{W}$ then $u \equiv x \pmod{W}$.

   Solution
   Reflexivity: $v + 0 = v = v$ and $0 \in W$, so $v \equiv v(\text{mod } W)$.
   Symmetry: $u \equiv v(\text{mod } W) \to \exists w$, $w \in W$ such that $u + w = v \to u = v - w \to v - w = u \to v \equiv u(\text{mod } W)$ because $-w \in W$
   Transitivity: $u \equiv v(\text{mod } W) \to \exists w$, such that $u + w_1 = v$. Same for $v \equiv x$, there $\exists w_2 \in W$ such that $v + w_2 = x$. And since $w_1 + w_2 \in W$(W is closed under addition), $u + w_1 + w_2 = v + w_2 = x$, so $u \equiv x(\text{mod } W)$.

7. Give an example of three subspaces Y_1, Y_2, and Y_3 in $\mathbb{R}^2$ such that $Y_1 + Y_2 + Y_3 = \mathbb{R}^2$ and $Y_i \cap Y_j = \{\mathbf{0}\}$ for all $i \neq j$, but $\mathbb{R}^2$ is not the direct sum of Y_1, Y_2, and Y_3. 2/3

: It's unclear what this means and how it gives you the result.

Solution

Take $Y_1 = span((0,1)) \quad Y_2 = span((1,0)) \quad Y_3 = span((1,1))$ The pairwise intersection is $\{0\}$, their sum is $\mathbb{R}^2$ but their direct sum isn't $\mathbb{R}^2$, because their intersection isn't trivial.

8. Prove that if V is a vector space and $W \subseteq V$ is a subspace, then W has a *complement*: 3/3
a subspace $U \subseteq V$ such that $V = W \oplus U$. Hint: V/W has a basis; lift it back to V.

Solution Take the basis for V/W. Take one element(non-zero) from each basis(coset) and the union of these with the set of the basis for W will be a basis for V.

Proof: Everytime you take an elmenent from the cosets, apply gram-schidt(Assume you started with an orthogonal set of basis for W) process to it, and then you will end up with an orthogonal set of basis. This is because you will never get a zero during this process, if you got zero during this process, that would imply that the vector you used is a linear combitionaion of the other ones and therefore couldn't be a basis for V/W.

After you have made this set of basis the elements that are not a basis for W, will be a a basis for U. Their direct sum will also be V, since they have no intersection.

9. For vectors $\mathbf{x} = (1, 2i, 1+i)$ and $\mathbf{y} = (i, 2-i, 3)$, compute

(a) $\langle \mathbf{x}, \mathbf{y} \rangle$, $||\mathbf{x}||^2$, $||\mathbf{y}||^2$, and $||\mathbf{y}||$; 3/3

(b) $\langle 3\mathbf{x}, 2i\mathbf{y} \rangle$ and $\langle 2\mathbf{x}, i\mathbf{x} + 2\mathbf{y} \rangle$; 3/3

(c) $||\mathbf{x} + 2\mathbf{y}||$. [Use parts (a) and (b) for this.] 3/3

Solution

(a) $\langle \mathbf{x}, \mathbf{y} \rangle = \bar{y}^T x = (-i, 2+i, 3)^T \cdot (1, 2i, 1+i) = -i + 4i - 2 + 3 + 3i = 1 + 6i.$
$||\vec{x}||^2 = \bar{x}^T x = (1, -2i, 1-i)^T \cdot (1, 2i, 1+i) = 1 + 4 + 1 - i + i + 1 = 7$
$||\vec{y}||^2 = \bar{y}^T y = (-i, 2+i, 3)^T \cdot (i, 2-i, 3) = 1 + 4 + 1 + 9 = 15$
$||y|| = \sqrt{15}$

(b) $\langle 3\mathbf{x}, 2i\mathbf{y} \rangle = -6i \langle \mathbf{x}, \mathbf{y} \rangle = -6i + 36$
$\langle 2\mathbf{x}, i\mathbf{x} + 2\mathbf{y} \rangle = 2 \cdot (-i \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle) = 2 \cdot (-7i + 2 + 12i) = 10i + 4$

(c)

$$\begin{aligned}
||\mathbf{x} + 2\mathbf{y}|| &= \sqrt{\langle x + 2y, x + 2y \rangle} \\
&= \sqrt{\langle x, x \rangle + \langle x, 2y \rangle + \langle 2y, x \rangle + \langle 2y, 2y \rangle} \\
&= \sqrt{||x||^2 + ||2y||^2 + 2\langle x, y \rangle + 2\langle y, x \rangle} \\
&= \sqrt{7 + 60 + 4} \\
&= \sqrt{71}
\end{aligned}$$

10. Prove that for vectors in an inner product space, $||\mathbf{x} \pm \mathbf{y}||^2 = ||\mathbf{x}||^2 + ||\mathbf{y}||^2 \pm 2\mathrm{Re}\langle \mathbf{x}, \mathbf{y} \rangle$. 1/3

Solution

$\langle \mathbf{x} \pm \mathbf{y}, \mathbf{x} \pm \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \pm \mathbf{y} \rangle \pm \langle \mathbf{y}, \mathbf{x} \pm \mathbf{y} \rangle = (\langle \mathbf{x}, \mathbf{x} \rangle \pm \langle \mathbf{x}, \mathbf{y} \rangle) \pm (\langle \mathbf{y}, \mathbf{x} \rangle \pm \langle \mathbf{y}, \mathbf{y} \rangle)$
$||\mathbf{x}||^2 + ||\mathbf{y}||^2 \pm \langle \mathbf{x}, \mathbf{y} \rangle \pm \overline{\langle \mathbf{x}, \mathbf{y} \rangle} = ||\mathbf{x}||^2 + ||\mathbf{y}||^2 \pm 2Re(\langle \mathbf{x}, \mathbf{y} \rangle)$

: This can't be right because $(x, y) - \overline{(x, y)} = 2\,\mathrm{im}(x, y)$

11. For any $m \times n$ complex matrix A, prove that $\ker(A^*A) = \ker(A)$. 3/3

Solution

Let $\mathbf{x} \in F^n$ such that $\mathbf{x} \in ker(A) \to A^*A\mathbf{x} = A^*(0) = 0 \to \mathbf{x} \in ker(A^*A)$
Let $\mathbf{x} \in F^n$ such that $A^*Ax = 0 \to (\mathbf{x}^*A^*)A\mathbf{x} = \langle A\mathbf{x}, A\mathbf{x}\rangle = 0 \to A\mathbf{x} = 0 \to \mathbf{x} \in ker(A)$.

12. Prove that if P is self-ajoint (that is, $P^* = P$) and idempotent (that is, $P^2 = P$) then 3/3
P is the matrix for an orthogonal projection.

Solution Defintion(I was not given any defintion) assumed:- P is orthogonal projection iff $P\mathbf{x}\mathbf{x} \perp P\mathbf{x}$
$\langle P\mathbf{x} - \mathbf{x}, P\mathbf{x}\rangle = \langle P^*(P\mathbf{x} - \mathbf{x}), \mathbf{x}\rangle = \langle (P^2\mathbf{x} - P\mathbf{x}), \mathbf{x}\rangle = \langle P\mathbf{x} - P\mathbf{x}, \mathbf{x}\rangle = \langle 0, \mathbf{x}\rangle = 0$
Therefore P is orthogonal projection.

13. If V is a vector space over $\mathbb{C}$ of dimension n, then it is also a vector space over $\mathbb{R}$ of dimension $2n$. (If this isn't clear to you, then write down a proof.) Given a Hermitian inner product $\langle \mathbf{x}, \mathbf{y} \rangle$ on V as a complex vector space, show that the real part $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{R}} = \mathrm{Re}(\langle \mathbf{x}, \mathbf{y} \rangle)$ is an inner product on V as a real vector space. 2/3

Solution

We can basically ask the question, if you only take the real part of the complex Hermitian inner product will it give you a valid inner product for the real vector space. So we just need to check if it satisfies the properties of linearity, conjugate symmetry, nonegativity and nondegeneracy.

3 and 4 are fine, but you should give a demonstration for 1 and 2.

1. Linearity
This property is simply true because the linearity is satisfied in complex inner product, and if you take the real part of it, it also has to be linear otherwise it wouldn't be linear in the compolex inner product to begin with, or you could make a map from from the norm to the real numbers, and since the norm is linear and this function is linear, their composition is also linear.

Norm is only $\mathbb{R}$-linear so this second justification doesn't work. And it's not as if this $\mathbb{R}$ restriction is given by such a composition.

2.Conjugate Symmetry
Just like Linearity, when you only take the real part, there has to be symmetry otherwise it wouldn't be symmetrical in the complex inner product anyway.

3.Nonnegativity
The product is already real in the complex inner product.

4.Nondegeneracy
The product is already real and nondegenerate in the complex inner product.

14. Find all possible Jordan forms of linear transformations with characteristic polynomial 3/3
$(t-1)^2(t+2)^2$.

Solution

$$\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & -2 & 0\\0 & 0 & 0 & -2\end{bmatrix} \begin{bmatrix}1 & 1 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & -2 & 0\\0 & 0 & 0 & -2\end{bmatrix} \begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & -2 & 1\\0 & 0 & 0 & -2\end{bmatrix} \begin{bmatrix}1 & 1 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & -2 & 1\\0 & 0 & 0 & -2\end{bmatrix}$$

Including permutaions on Jordan blocks.

15. Find all possible Jordan forms of linear transformations with characteristic polynomial $(t-2)^3(t+1)$ and minimal polynomial $(t-2)^2(t+1)$. 3/3

$$\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$ Up to permutation of the Jordan blocks.

16. How many similarity classes are there for 3×3 matrices whose only eigenvalues are -3 and 4? 3/3

Solution The answer is 4. There are at least 2 eigenvectors, with eigenvalues -3 and 4. The third one can be in either and the geomtric multiplicies of the two eigenvalues can be either 1 or 2 for each, giving a total of 4 equivalence classes.

17. Prove or disprove: two $n \times n$ matrices are similar if and only if they have the same characteristic polynomial and minimal polynomial.

$2/3$

Your counterexample is flawed because geometric multiplicity 3 is incompatible with $4, 2, 2$.

Solution

Unfortunately, two matrix having the same characteristic and minimal polynomial, doesn't gurantee that they are similar, it is a requirement but it is not a suffiecient requirement.

As an example consider a matrix whose eigenvalue has geomtric multiplicity 3, algebraic multiplicty 8(I didn't write the matrix out because it was 128 numbers), so that you can get two matrices, such that one has 3 jordan blocks of size 4,2,2 and the other has 3 jordan blocks of size 4,3,1. They have the same minimal polynomial and characteristic polynomial, but their jordan blocks are different and therefore can't be the similar.